Grade 6 Accelerated Name____________________
Unit 14 Date____________________
Lesson 1 Practice Period____________________

A local town has a sign ordinance that requires all store signs be no more than 11 feet from the ground.

1. Write an inequality to represent the height, h, that store signs can be.

2. Represent the inequality on a number line.

3. List two examples of possible heights, in feet, of the store signs.

4. Give one example of a height that is true for the inequality but does not make sense in the real world.

The maximum height for a rider on a specific theme park ride is 6.5 feet.

5. If h represents the height of the rider, write an inequality to describe all possible values of h.

6. Represent the inequality on a number line.

7. Does the inequality that represents the situation include 0 as a possible height? Why or why not?

8. Explain why a negative number does not make sense in this context.

A student needs to obtain a minimum of 100 volunteer hours in order to achieve the maximum Florida Scholarship Award.

9. If v represents the number of volunteer hours that a student completes, write an inequality that represents the number of volunteer hours completed.

10. Represent the inequality on a number line.

Grade 6 Accelerated
Unit 14
Lesson 2 Practice

Name____________________
Date____________________
Period____________________

Solve.

1. $\frac{1}{5} \cdot b = \frac{7}{10}$

2. $4 = g - \frac{1}{4}$

3. $p \div \frac{14}{15} = 2$

4. $\frac{1}{2} = 4\frac{1}{2} + t$

5. $c + 1\frac{1}{4} = 2\frac{1}{6}$

6. $\frac{1}{4}r = \frac{5}{36}$

7. In preparation for a party, Suzanne filled four equal sized candy jars with $3\frac{2}{3}$ cups of various candies. How many cups of candy did Suzanne use?

8. After lighting a candle, $\frac{3}{8}$ inches of it were burned before it was put out, leaving $2\frac{1}{6}$ inches of the candle remaining. How long was the candle before it was lit?

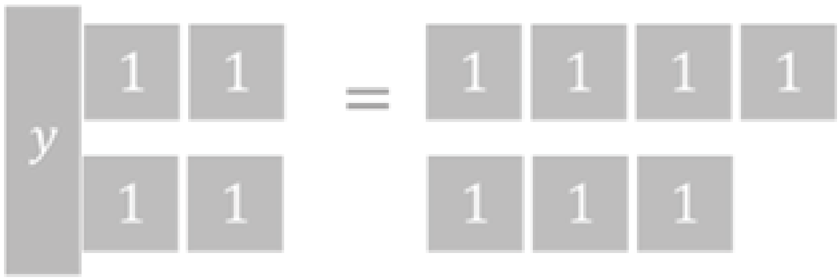
9. A 20 ounce soda bottle has $12\frac{2}{5}$ ounces remaining. How much soda was removed from the bottle?

10. Peter bought a new premium bike helmet for $\$81.77$ from Store A, which was $\$9.24$ cheaper than the same helmet on sale at Store B. What was the price of the helmet at Store B?

Grade 6 Accelerated
Unit 14
Lesson 3 Practice

Name____________________
Date____________________
Period____________________

For questions 1-5, use the algebra tiles shown.



1. What expression represents the left side of the equation?

2. What expression represents the right side of the equation?

3. Write the algebraic representation of the equation.

4. What operation would be used to solve the equation?

5. What value of y makes the equation true?

Solve.

6. $5 + x = 12$

7. $9 = -16 + h$

8. $y + 10 = -113$

9. $y - 5 = -24$

10. $-9 + x = 13$

Grade 6 Accelerated
Unit 14
Lesson 4 Practice

Name____________________
Date____________________
Period____________________

Solve.

1. $75 = 12 + x$

2. $x - 3 = 23$

3. $92 = -81 + h$

4. $-103 = g + 7$

5. $y - 10 = 46$

6. $-19 + u = 31$

7. $y - 66 = 88$

8. $-8 + x = 51$

9. $-13 + x = -27$

10. $-41 + x = -903$

Grade 6 Accelerated
Unit 14
Lesson 5 Practice

Name____________________
Date____________________
Period____________________

Write and solve an algebraic equation for each of the following.

1. Rhonda ate 8 peanut M&M's, which is 6 less than her daughter, Diana ate. How many peanut M&M's did Diana eat?

2. Cathy reviewed her monthly spending and found that she reduced her monthly spending on coffee by $\$32$ from the prior month. If her monthly spending on coffee was $\$108$, how much did she spend on coffee in the prior month?

3. Marcus gained 350 new followers on social media after his article was published. If he now has $3{,}425$ followers, how many did he have before the article was published?

4. Fischer has a goal of making $\$120$ in tips before the end of his shift. If he already has earned $\$91$, how much more does he need in order to meet his goal?

5. Darlene collected 28 cans for a food drive, which was 12 less than the prior day. How many cans did she collect the prior day?

6. Bruno earned $19.50 for washing the family car, bringing his total amount of savings to $66.20. How much did Bruno already have saved before washing the car?

7. Xin ran 18.25 miles less than her friend Kelly ran last week. If Xin ran 22.9 miles, how many miles did Kelley run?

8. After paying cash for dinner, Daniel received $9.91 back as change. If the total of the bill was $40.09, how much did Daniel give the waiter to pay the bill?

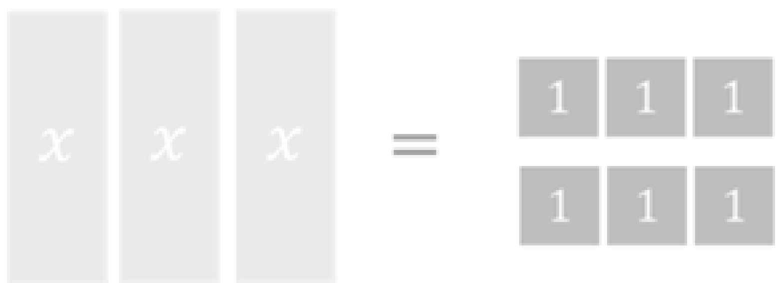
9. Alyssa beat her high score by scoring 9,241 points more than her previous high score. If her original high score was 104,560 points, what is her new high score?

10. A streaming service recently lost 4,500 subscribers. If they originally had 92,000 subscribers, how many subscribers do they have now?

Grade 6 Accelerated
Unit 14
Lesson 6 Practice

Name____________________
Date____________________
Period____________________

For questions 1-5, use the algebra tiles shown.



1. What expression represents the left side of the equation?

2. What expression represents the right side of the equation?

3. Write the algebraic representation of the equation.

4. What operation would be used to solve the equation?

5. What value of x makes the equation true?

6. $-4x = 16$

7. $-\frac{y}{3} = 19$

8. $-22 = 11x$

9. $\frac{b}{10} = 18$

10. $16 = \frac{f}{14}$

Grade 6 Accelerated
Unit 14
Lesson 7 Practice

Name____________________
Date____________________
Period____________________

Solve.

1. $-30 = \frac{j}{-6}$

2. $-6x = 66$

3. $-\frac{h}{9} = 81$

4. $10 = 2m$

5. $16 = 4x$

6. $-12 = \frac{y}{3}$

7. $\frac{r}{6} = 13$

8. $12p = -60$

9. $3 = \frac{y}{-2}$

10. $42 = -x$

Grade 6 Accelerated
Unit 14
Lesson 8 Practice

Name____________________
Date____________________
Period____________________

Write and solve an algebraic equation for each of the following.

1. At a restaurant, 9 friends decided to split the bill evenly. If the total bill was $\$170.28$, how much would each friend pay?

2. A bride-to-be is stuffing wedding favor bags using a random mix of her 500 pieces of party favors to split evenly among her guests that attend the wedding. If each guest received 20 pieces, how many guests attended the wedding?

3. Jake has a budget of $\$200$ for groceries each month. How much is Jake's budget for groceries for an entire year?

4. During a week, Brodie earned $\$58.40$ for walking dogs. If Brodie walked 8 dogs, how much did he charge for each walk?

5. At a local arcade, the number of tickets needed for a stuffed bear is $20{,}000$ tickets. If Darnell earns 500 tickets each time he visits the arcade, how many visits will it take to get the stuffed bear?

6. Deborah has 80 books that she wants to put on her new 5-shelf bookshelf. If she distributes the books evenly to each shelf, how many books will be on each shelf?

7. A recipe for pancakes calls for 2 cups of water for each batch. If someone wants to make 3 batches, how many cups of water will be needed?

8. A car manufacturer offers 9 different car models, each with 6 color options. How many car choices would a customer buying from that manufacturer have?

9. A shoe storage bin can hold 8 pairs of shoes. Kiera purchased 4 bins, how many pairs of shoes can she store?

10. Minnie is selling lemonade to raise money. At the end of the day, she raised $\$10.60$. If each cup sold for $\$0.25$, how many cups of lemonade did Minnie sell?

Grade 6 Accelerated Name____________________
Unit 14 Date____________________
Lesson 9 Practice Period____________________

1. For each inequality, write three numbers that could represent the solution set.

   a. $\frac{1}{2} < p$

   b. $-3 \leq g$

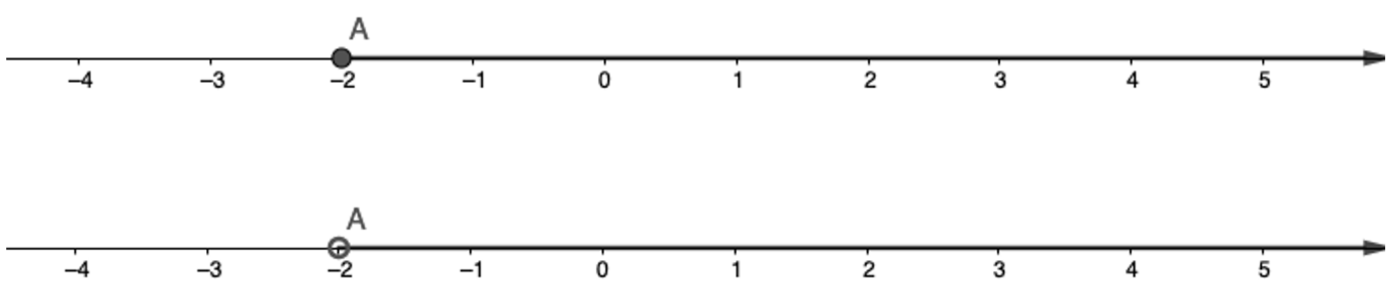
   c. $w < 10$

2. You have to be at least 18 years old to see the new scary movie at the movie theater. Write an inequality to represent the ages that one can be to see the scary movie in the theater. Let a represent the age.

3. Graph the solution $16 \leq x$.

4. Stephanie states that the numbers $(6, 8, 10)$ could be in the solution set for the inequality $6 \geq t$. Explain why Stephanie's answer is accurate.

5. Compare and contrast the two number lines.



6. Use the inequality $-28.5 < v - 16$ to complete the following:

a. Find the solution set by isolating the variable. Show your work.

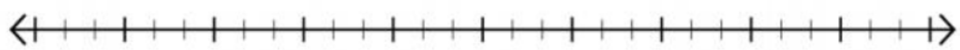
b. Graph the solution set on the number line.

c. Use substitution to test a value or values in order to verify that the solution and the graph are correct.

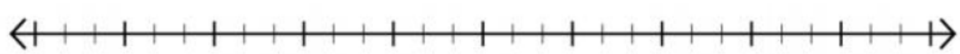
7. How do you isolate a variable when solving an inequality?

Solve each inequality. Then, graph its solution set.

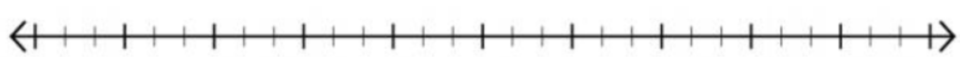
8. $6 > z - 6$



9. $47\frac{3}{4} \geq w + 22$



10. $x + 8.2 < 18$



Grade 6 Accelerated Name____________________
Unit 14 Date____________________
Lesson 10 Practice Period____________________

1. A great white pelican can hold about 700 cubic inches in its bill. After flying down to try to catch a fish, the pelican has 100 cubic inches of water in his mouth. The inequality $700 \leq 100 + w$ represents this situation, with w representing the amount of space that can still be filled in the pelican's bill. What is the maximum amount of water that can be added to the bill without spilling out?

2. Clay County is gearing up for their annual Math Field Day competition. They need to order boxed lunches for all the proctors. Chic-Fil-A charges them $\$6.75$ per box for lunch. The budget for the event lunches is $\$225$. The inequality $\$6.75x \leq \$225$ represents the situation, with x representing the number of box lunches that can be purchased from Chic-Fil-A. What is the maximum number of boxed lunches that can be purchased?

3. The band department is raising money to purchase new uniforms by selling Mums for senior night. The team earns $\$0.75$ for each flower sold. They hope to raise at least $\$250$ from this fundraiser. This situation is represented by the inequality $\$0.75x \geq \$250$. What is the minimum number of flowers that the band must sell to reach their fundraising goal?

4. Maximum capacity at Busch Gardens, Tampa is $33{,}000$ people. The inequality $b \leq 33{,}000$ represents this situation, with b representing the number of patrons at the park. Give an example of a solution to the inequality that is not valid in this context.

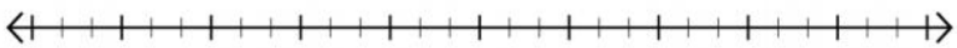
5. Hilary is buying cupcakes from Publix to share with her 26 classmates for her birthday. The store has 18 vanilla cupcakes on the shelf. She wants to buy the remaining chocolate for the remaining cupcakes. The inequality $18 + c \geq 26$ represents the situation, with c standing for the number of cupcakes that she still needs to buy. Give an example of a solution to the inequality that is not valid in this context.

6. Find the solution to both inequalities using the division property of inequality.

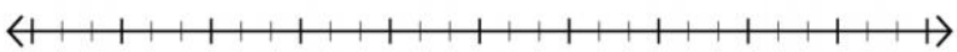
$7y < -105$	$-7y < -105$

Solve each inequality. Then, graph its solution set.

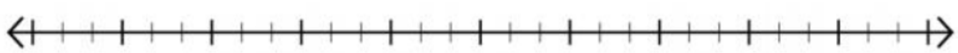
7. $-\frac{c}{2} < -1.5$



8. $-4x > 52$



9. $5x \leq -55$



10. How is the process of solving an inequality with a negative coefficient similar to the process of solving an inequality without a negative coefficient?